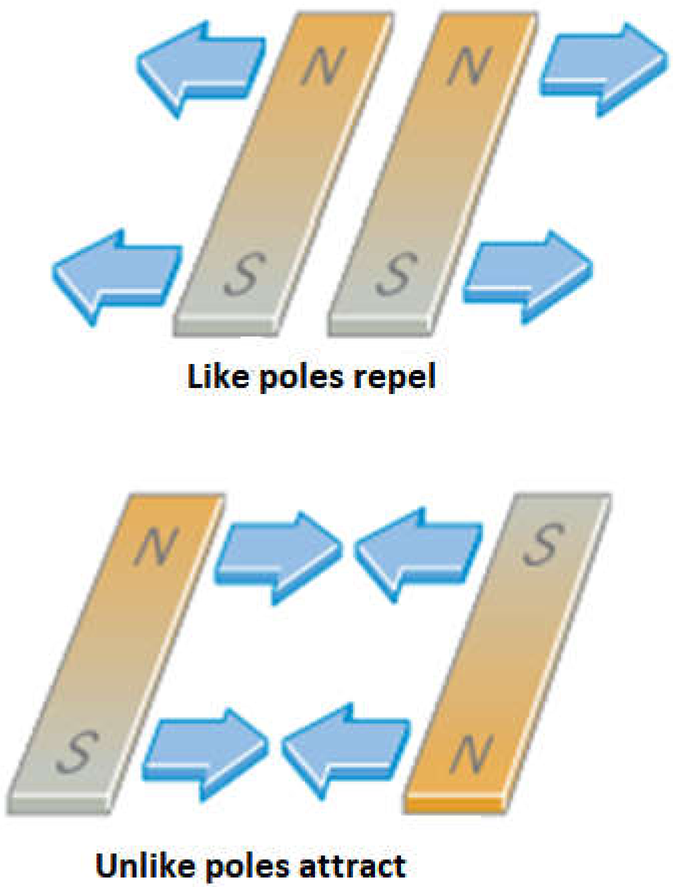
Chapter 19 Magnetism

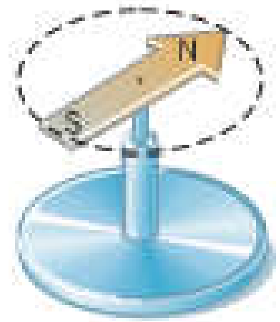
In this chapter we will study the following:

- ✓ Permanent magnets as a source of magnetic filed and field lines
- ✓Earth magnetic field
- ✓ Electric current as a source of magnetic field
- ✓Magnetic force on a moving charged particle
- ✓Magnetic force on current-carrying wire
- ✓Magnetic dipoles and potential energy of dipole-magnetic field system
- ✓ Magnetic torque on a dipole subject to magnetic field
- ✓Magnetic moment of an orbiting charge and its relation with angular momentum (Application: Bohr model)
- ✓ Magnetic field produced by currents of :
  - ❑ Long straight wire
  - ❑ Circular loop
  - ❑ Long Solenoid
- ✓ Magnetic force between two parallel carrying wires

Chapter 19 Magnetism

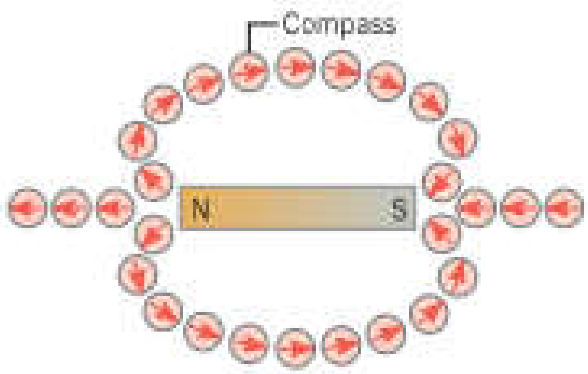
19.1 Magnetic Fields:



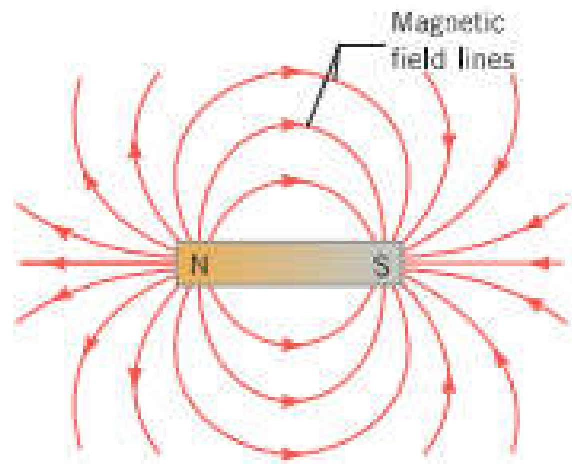


The needle of a compass is permanent magnet that has a north magnet pole (N) at one end and a south magnetic pole (S) at the other

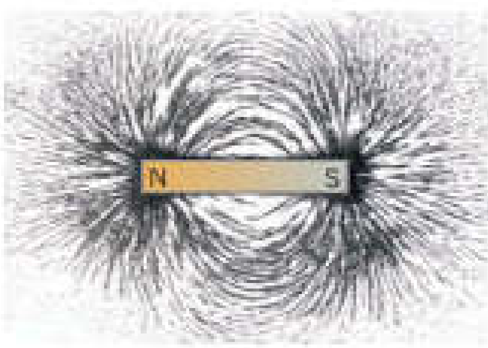
Mapping a Magnetic Field B:



At any location in the vicinity of a magnet, the north pole (the arrowhead in this drawing) of a small compass needle points in the direction of the magnetic field at that location.

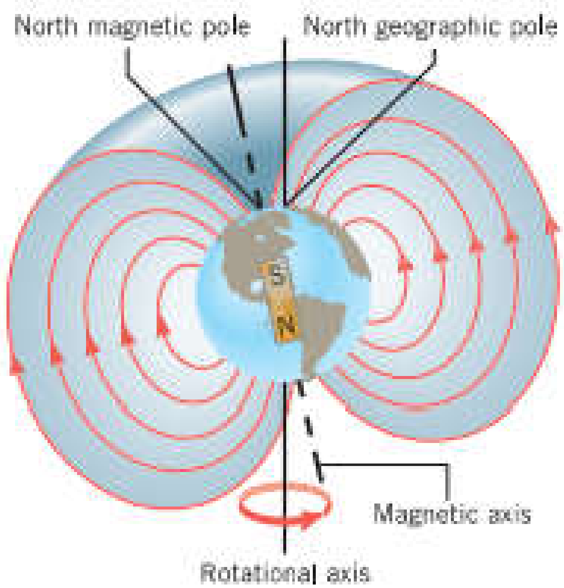


The magnetic field lines



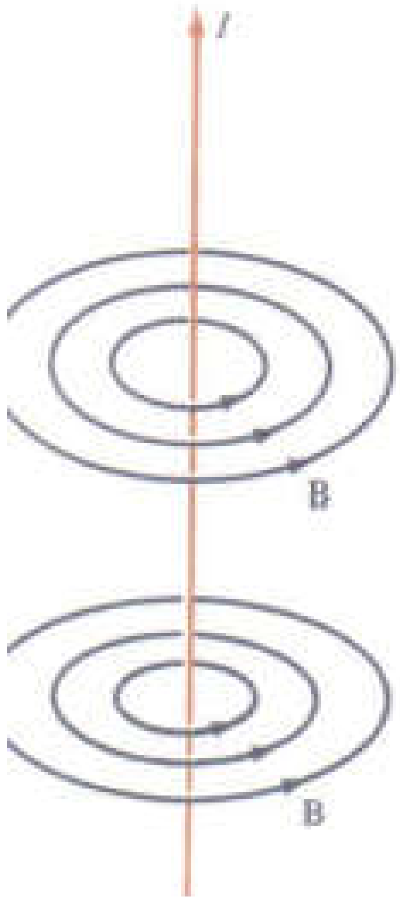
The pattern of Iron filings in the vicinity of bar magnet

Earth Magnetic Field

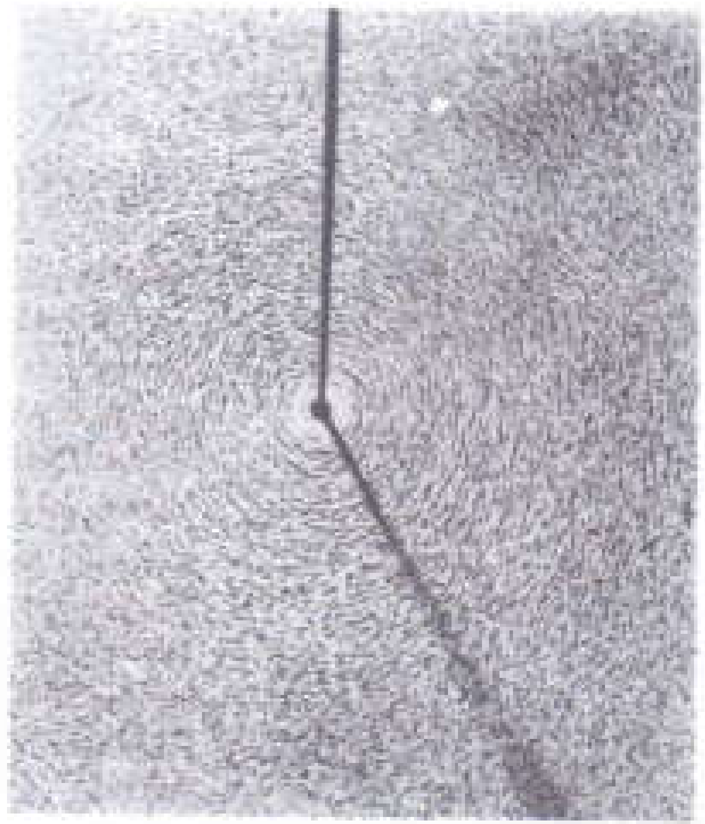


The earth behaves magnetically almost as if a bar magnet were located near its center. The axis of this fictitious bar magnet does not coincide with the earth's rotational axis; the two axes are currently about 11.5° apart.

Magnetic Field Due to a Long Straight wire of current I



(a)

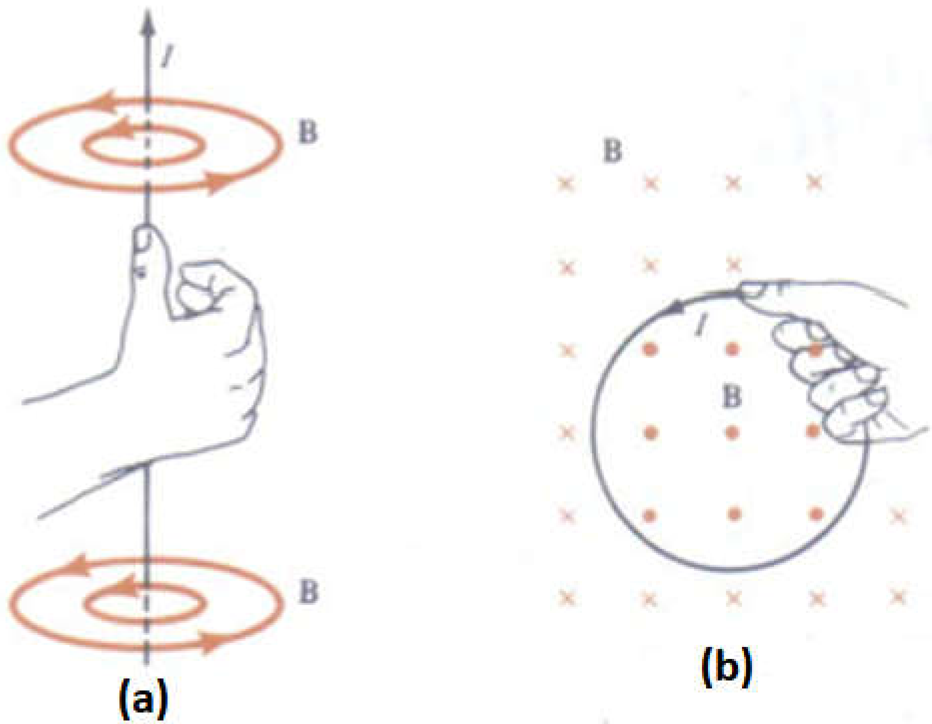


(b)

(a) Magnetic field lines and (b) iron filing patterns due to a current in a long straight wire

Direction of B

(a) Long straight wire carrying current I
(b) Current loop carrying current I

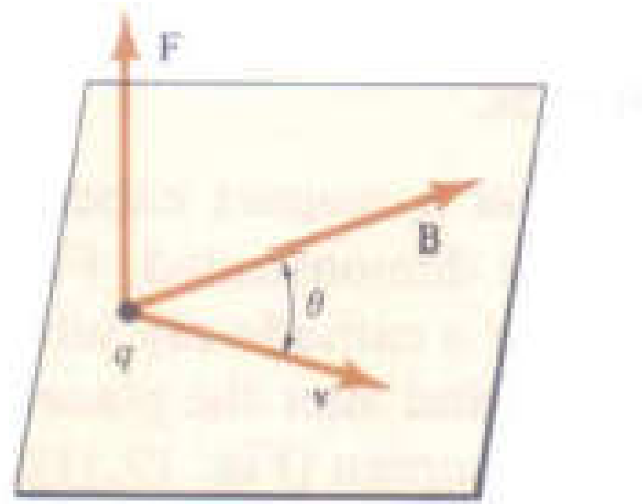


(a)With the right thumb along the direction of the current, the fingers curl in the direction of the magnetic field (b)Inside the loop, B points out of loop: outside the loop, B points into the page

19.2 The Magnetic Force on a Moving Charge

The magnetic force on +q is given by:

$$\vec{F} = q\vec{V} \times \vec{B}$$

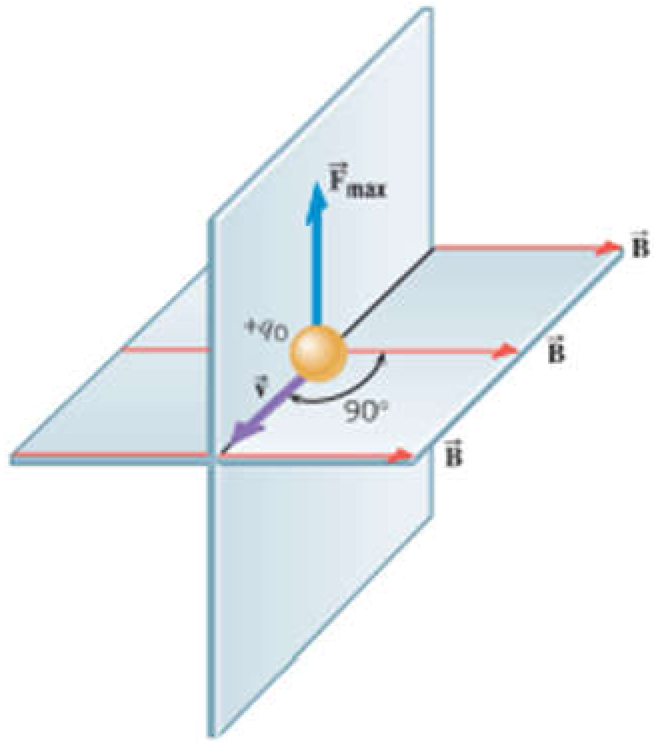


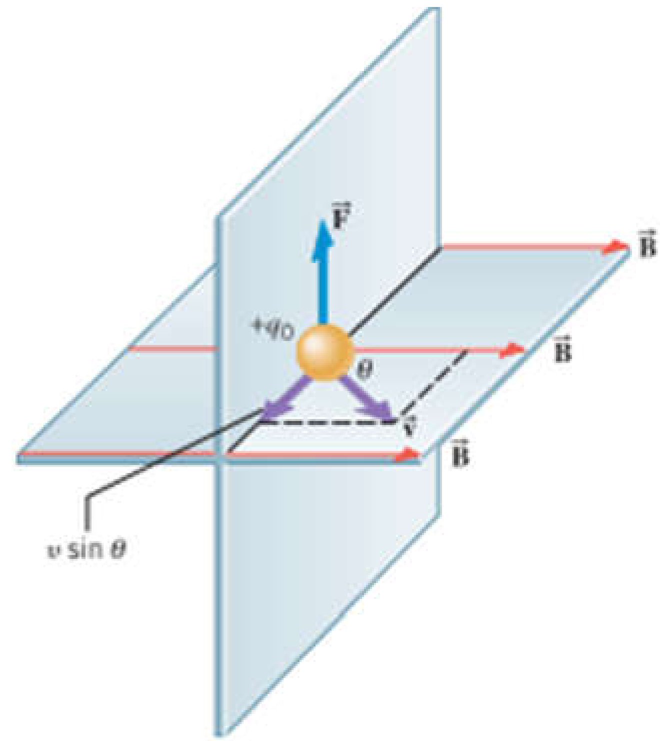
The magnetic force is:

$$F = |q| V\, B \sin(\theta)$$

- **For negative charge the direction of the force is opposite to that of positive charge**

$$\text{SI Units of Magnetic Field: } \frac{\text{Newton Second}}{\text{Coulomb meter}} = 1 \text{ tesla (T)}$$





The charge experiences a maximum force F_{max} when the charge moves perpendicular to the field B

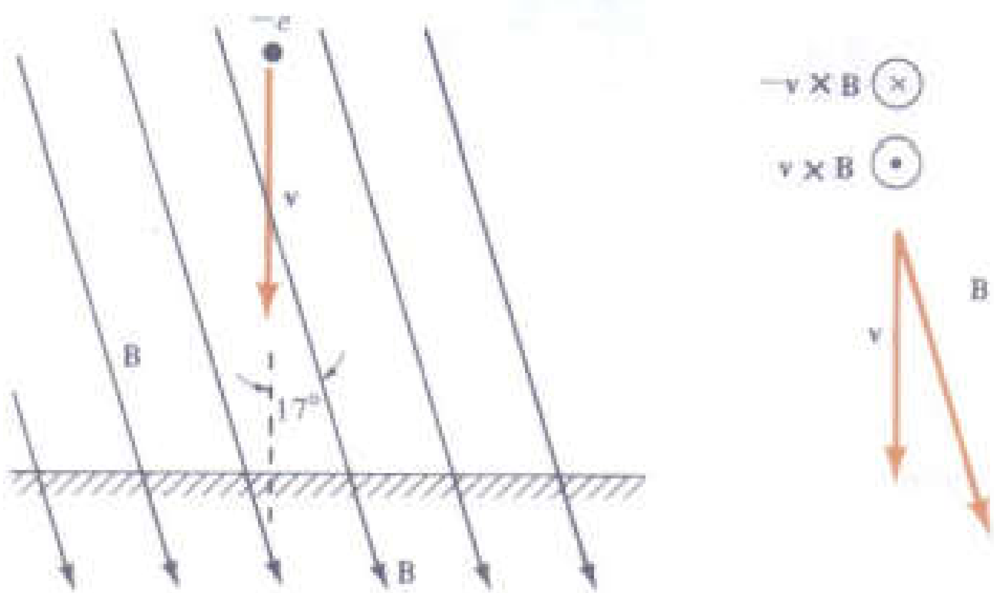
If the charge travels at an angle theta with respect to B, only the velocity component perpendicular to B gives rise to a magnetic force F, which is smaller than Fmax. This is v sinθ

- **Two conditions must be met for charge to experience a magnetic force when placed in a magnetic field**

1. **Charge must be *moving*, for no magnetic force acts on a stationary object**
2. **The velocity of the moving charge must have a *component* that is *perpendicular* to the direction of the magnetic field.**

- A) no magnetic force acts on a charge moving with a velocity v that is parallel or antiparallel to a magnetic.
- B) the charge experiences a maximum force F_{max} when the charges moves perpendicular to the field
- C) if the charge travels at an angle θ with respect to B, only the velocity component perpendicular to B gives rise to a magnetic force F, which is smaller than Fmax. This component is v sin θ.

EXAMPLE 19.1: At Boston, Massachusetts, the magnetic Filed due to the earth is at 17° to vertical direction and has a magnitude of 5.8×10^{-5}T (see Fig.) (a) Find the force F on an electron moving straight down at 10^5ms/s. (b) Find the ratio of F to weight mg.



in
out

Solution:

(a) $q = -e = -1.6 \times 10^{-19} C \text{ and } \sin 17^0 = 0.292$

$$F = qvB\sin\theta = 1.6 \times 10^{-19} \times 5.8 \times 10^{-5} \times 0.292$$

$$= 2.71 \times 10^{-19} N$$

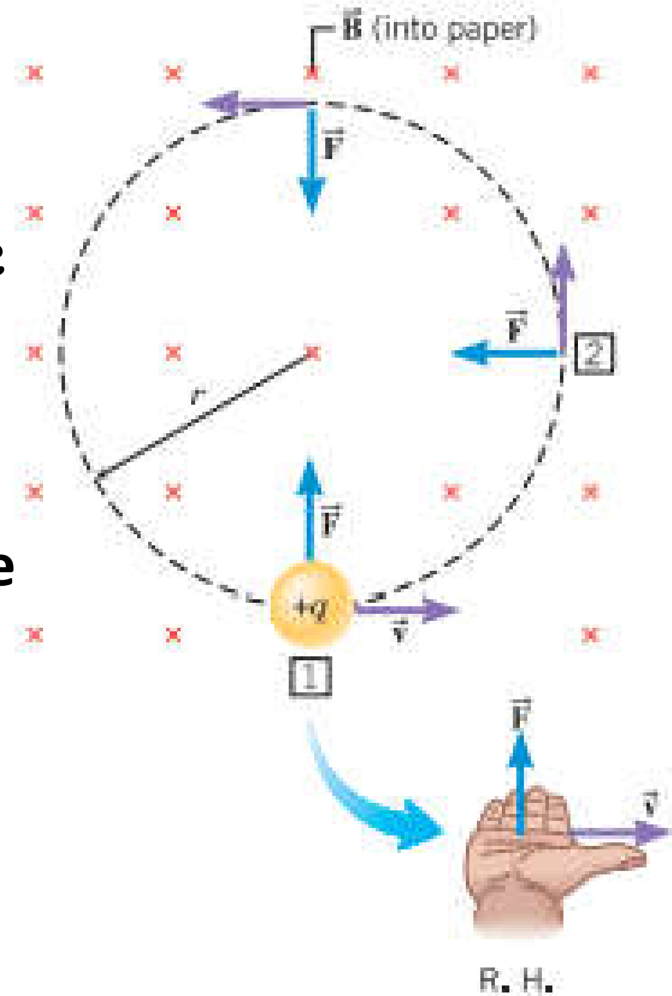
(b) *The mass of an electron is* $9.1 \times 10^{-31} kg$

$$\frac{F}{mg} = \frac{2.71 \times 10^{-19}}{9.1 \times 10^{-31} \times 9.8} = 3.04 \times 10^{10}$$

Direction of F is into the page

The Circular Trajectory

A positively charged particle is moving perpendicular to a constant magnetic filed. The magnetic force F causes the particle to move on a circular path. (R.H. = right hand)



The centripetal force equal to magnetic force

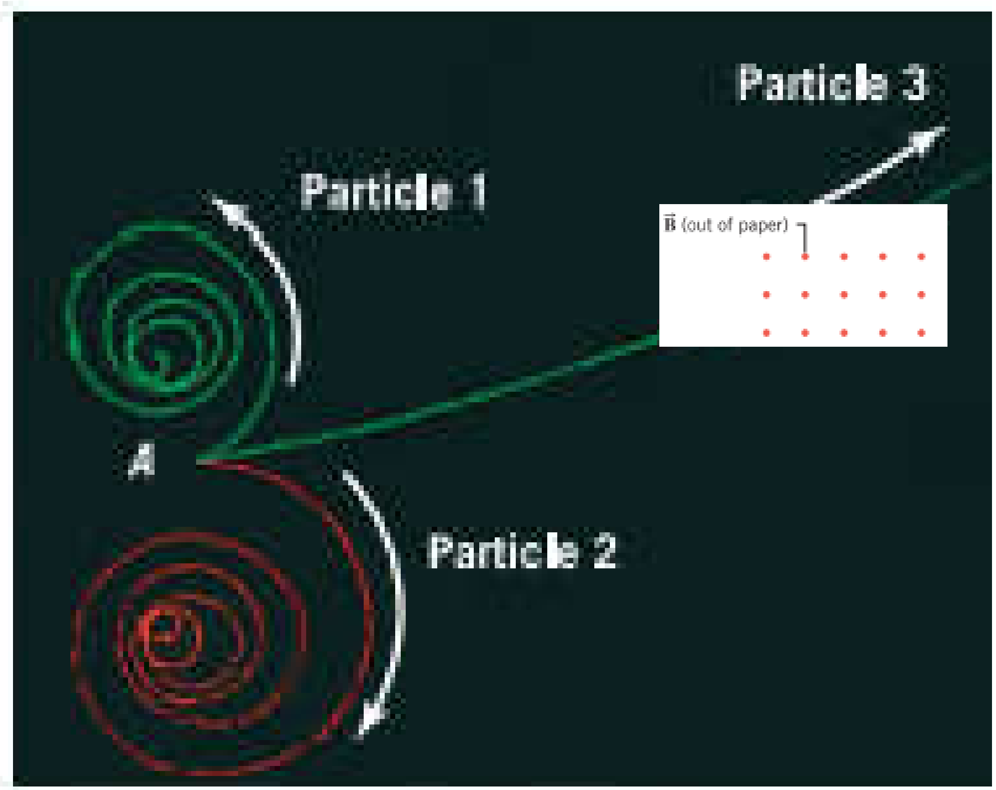
$$F_C = m\frac{v^2}{r} \Rightarrow |q|vB = m\frac{v^2}{r}$$

$$\Rightarrow r = \frac{mv}{|q|B}$$

This the radius of the orbit (circle)

EXAMPLE: Particle Tracks in a Bubble Chamber

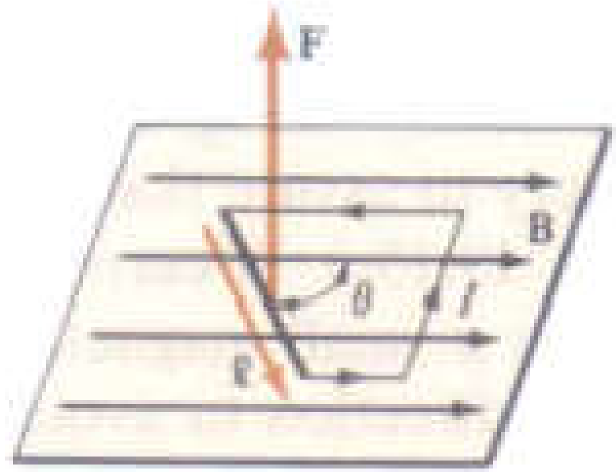
•Trace of three particles
•B is directed out
•Particle 1 and particle 3 move
On upward-curving tracks,
They are negative charges
•Particle 2 is positive



19.4 The Magnetic Force on a Current-Carrying Wire

The magnetic force on segment L carrying current I is given by:

$$\vec{F} = I\vec{L} \times \vec{B}$$



L is in the direction of current I

If the segment is at angle θ to the filed, the magnitude of the force is: $F = ILB\sin\theta$

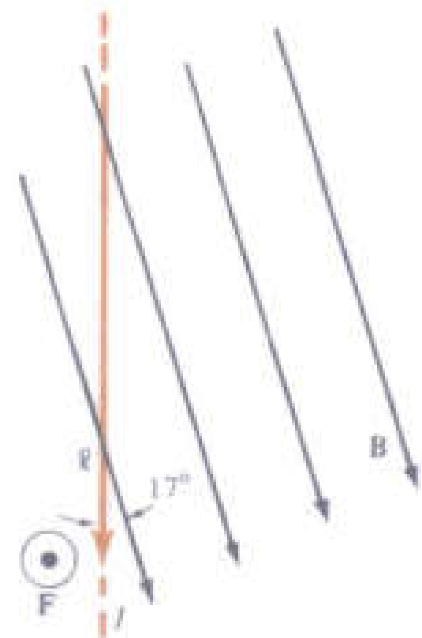
EXAMPLE 19.2: The magnetic field of the earth at Boston. Massachusetts, is at 17° to the vertical and has a Magnitude of 5.8×10^{-5}T (see Fig.). A vertical wire carries A current of 10A. Find the force on a 2m length of the Wire.

Solution:

The magntidue of the force is

$$F = ILB\sin\theta$$

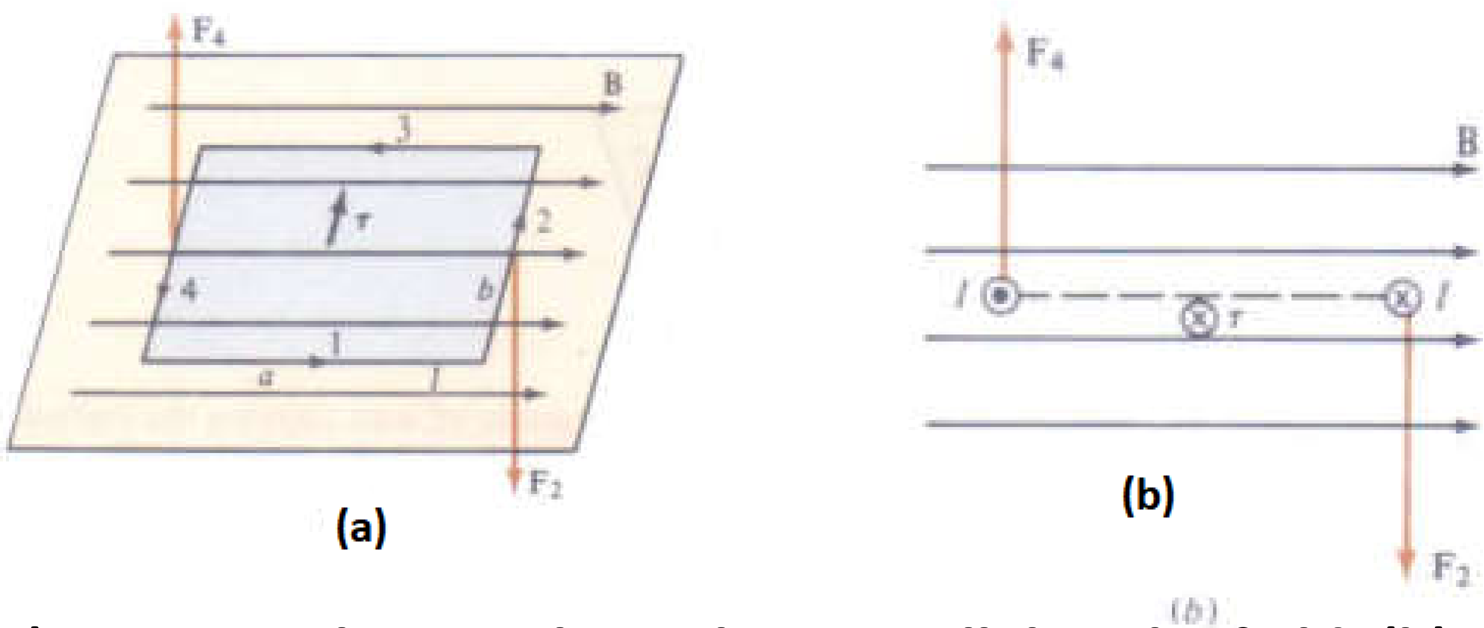
$$= 10\times2\times(5.8\times10^{-5})\sin17^{o}$$



The direction of the force is out of the page

19.5 Magnetic Dipoles

A loop of wire carrying a current behaves in a magnetic field much as an electric dipole does in an electric field and called a **magnetic dipole.**

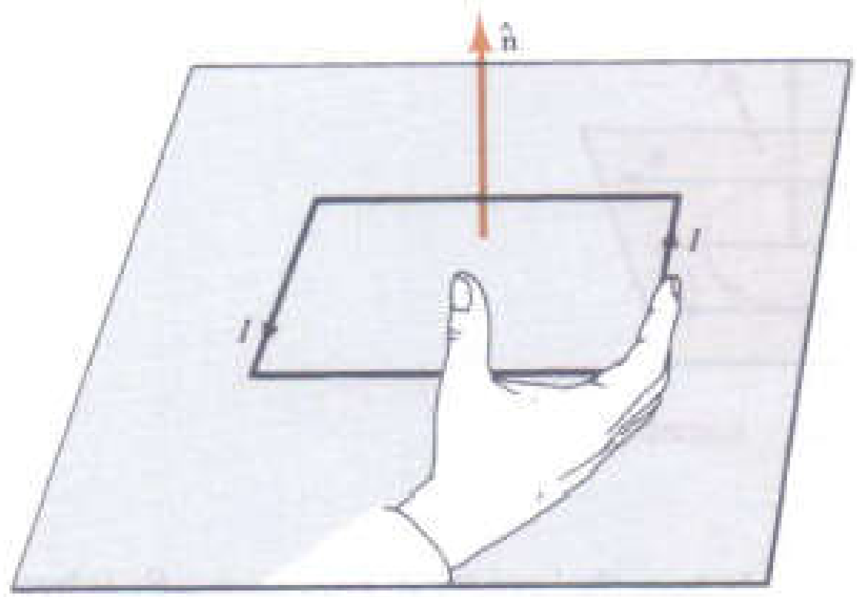


(a) A current loop with its plane parallel to the field. (b)The same loop seen in cross section. The dashed line indicates the plane of the loop

The magnetic torque is given by:

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\vec{\mu} = I A \hat{n}$$



The normal direction $\hat{n}$ indicated by the right thumb is upward when the fingers point along the current direction.

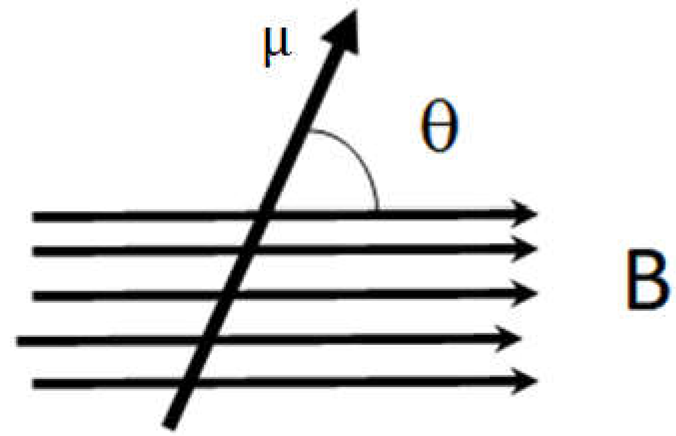
A is the area of loop

$\hat{n}$ is the normal to the area

Potential energy of magnetic dipole:

The magnetic potential energy of a magnetic dipole subject to Magnetic field B is:

$$u = -\vec{\mu} \cdot \vec{\boldsymbol{B}}$$



$$u = -\mu \, B \cos(\theta)$$

EXAMPLE 19.3: A circular turn of wire of radius 0.5m carries a current of 4A. (a)What is its magnetic field moment? (b)If the normal to the Loop is at 90° to a field of 2T, what is the magnitude of torque? (c)What is its Potential energy at this angle?

Solution:

(a) *The area of the loop is* πr^2

$$\mu = IA = I\pi r^2 = 4\times\pi\times 0.5^2 = 3.14\,Am^2$$

(b) $\tau = \mu B\sin\theta = 3.14\times 2\times \sin 90^{o} = 6.28N$

(c) $u = -\mu B\cos\theta = -\mu B\cos 90^{o} = 0$

$$bc\cos 90^{0} = 0$$

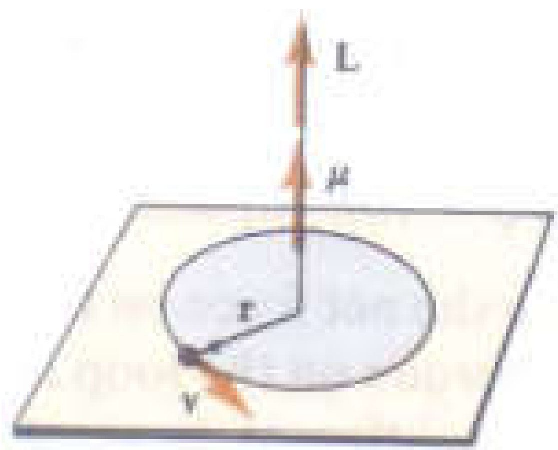
Magnetic Moment of an Orbiting Charge

Current produced by q is

$$I = \frac{q}{t} = \frac{q}{2\pi r/v} = \frac{qv}{2\pi r}$$

The magnetic moment of
The orbital charge is

$$\mu = IA = \frac{qv}{2\pi r}\pi r^2 = \frac{qvr}{2}$$



The magnetic moment due to an orbiting positive charge is parallel to its angular momentum.

The angular momentum L is

$$\vec{L} = \vec{r} \times \vec{P}, \quad \text{linear momentum } \vec{P} = m\vec{v}$$

$$\Lsh L = mvr \Rightarrow \vec{\mu} = \frac{q}{2m}\vec{L}$$

EXAMPLE 19.4: In the Bohr model of the hydrogen atom, The electron moves in a circular orbits for which the Angular momentum $L=nh/2\pi$, where n is integer and h is Planck's constant $h = 6.63\times10^{-34}$ Js. (a)Find the magnetic moment due to the orbital motion in the lowest (n=1)orbit. (b)Suppose an electron in the n=1 orbit has its moment parallel to a magnetic field of 10T. How much energy in electron volts must be Supplied to reverse the orientation of the moment?

Solution:

(a) With n=1, the angular momentum $L = h/2\pi$. Since $q = -e$, the magnetic moment has a magnitude

$$\mu = \frac{e}{2m}L = \frac{1.6\times10^{-19}\times6.63\times10^{-34}}{4\pi\times9.11\times10^{-31}}$$

$$= 9.26\times10^{-24}\,Am^2$$

(b) Since the potential energy of the magnetic dipole is $u = -\mu B\cos\theta$, the potential energy is $-\mu B$ when μ is along the field and $+\mu B$ when μ is opposite to the field. Hence, the energy needed to reverse the direction of the dipole is

$$u = +\mu B - (-\mu B) = 2\mu B$$

$$2\mu B = 2 \times 9.26 \times 10^{-24} \times 10$$

$$= 1.85 \times 10^{-22} J \times \frac{1ev}{1.6 \times 10^{-19} J}$$

$$= 1.16 \times 10^{-3} eV$$

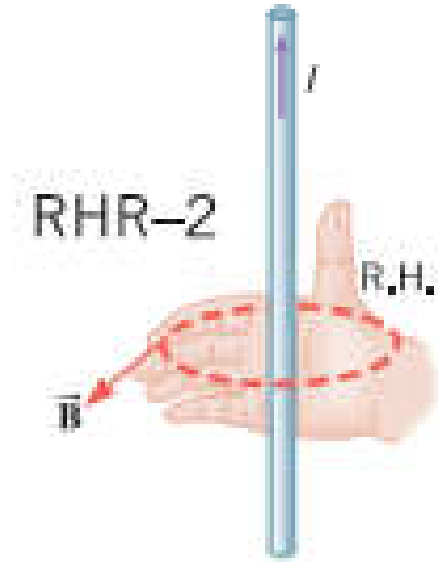
19.7 Magnetic Fields Produced by Currents

- **A Long Straight Wire**

Infinitely long, striaght wire

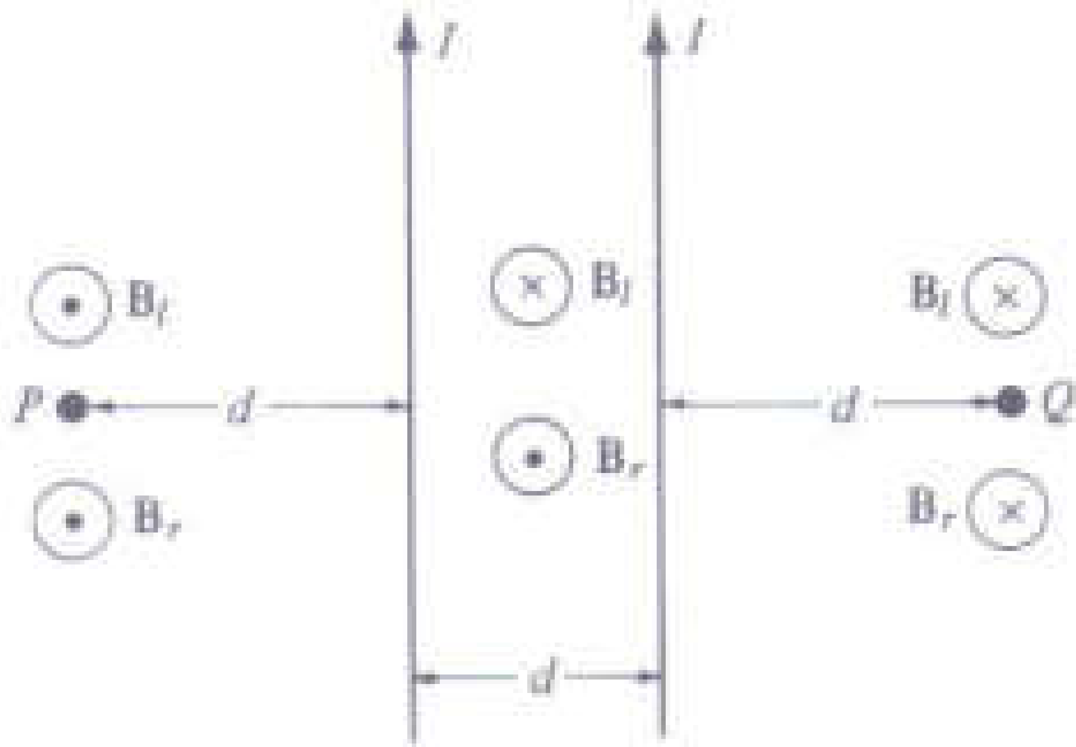
$$B = \frac{\mu_o I}{2\pi r}$$

$$\mu_o = 4\pi \times 10^{-7} T/m$$



r is the distance from the center of the wire to the field point

EXAMPLE 19.7: Two long straight wires a distance d apart Each carry a current I in the same direction (as in Figure) Find the net magnetic field (a) midway between the wires (b) at points P and Q in Fig.



Solution:

(a) At mid point $r = d/2$, $B_{left\ wire} = \mu_o I/(2\pi d/2)$ into the page

$B_{right\ wire} = \mu_o I/(2\pi d/2)$ out of the page

Therefore, the two fields are equal in magnitude and opposite in direction. So, the vector sum is **zero**

Hence, $B_{midway} = 0$

(b) At point P the fields add. So $B = B_{left\ wire} + B_{right\ wire}$

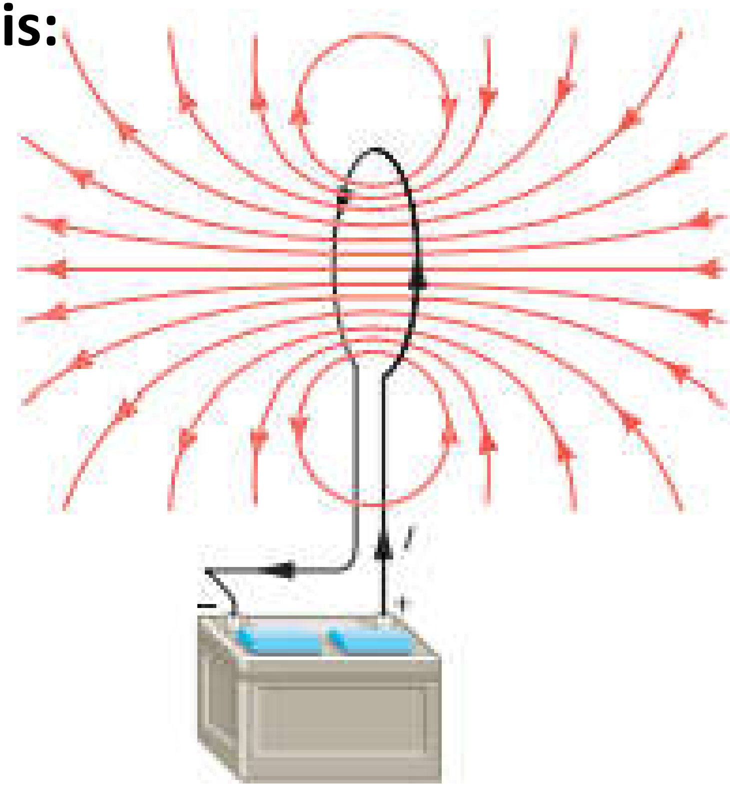
$B = \mu_o I/2\pi d + \mu_o I/(2\pi \times 2d) = 3\mu_o I/4\pi d$ out of the page.

The same magnitude of field at point Q but the direction of the file is into the page.

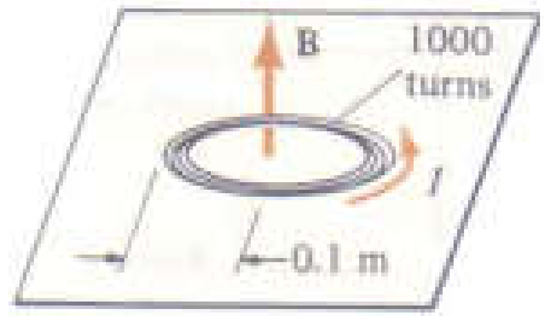
Magnetic Field of a Circular Current Loop:

The magnetic field B at the center of a loop of radius R and N-turns carrying current I is:

$$B = N\frac{\mu_o I}{2R}$$



Example:

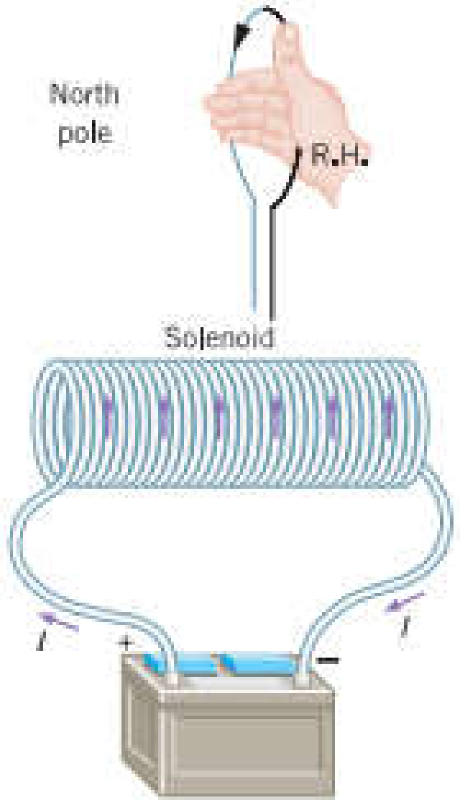


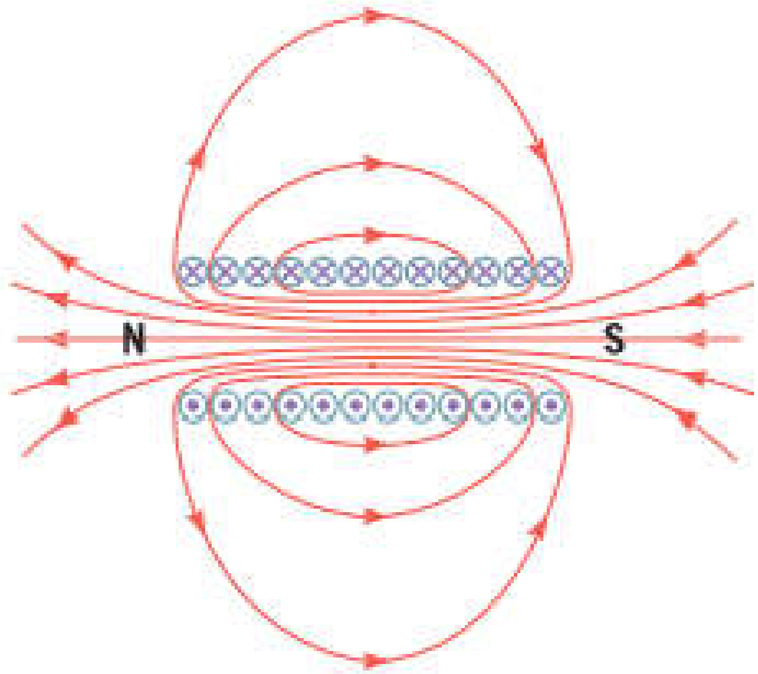
A coil consists of 1000 circular turns of
Thin wire with an average radius of 0.1
As shown aside. If the current in the coil
Is 10A. Find the magnetic field at its center due
to one turn (b) the entire coil

(a) $$B = N\frac{\mu_o I}{2R} = 1 \times \frac{4\pi \times 10^{-7} \times 10}{2 \times 0.1} = 6.28 \times 10^{-5} T$$

$$(b) B = N\frac{\mu_o I}{2R} = 1000 \times \frac{4\pi \times 10^{-7} \times 10}{2 \times 0.1} = 0.0628 T$$

Magnetic Field of a Long Solenoid:





A solenoid and cross sectional view of It, showing the magnetic lines and the North and south poles

B inside along Solenoid is:

$$B = \mu_o n I$$

n is the # of turns per unit length

I is the current

EXAMPLE 19.6: A person studying the magnetic bacteria Wishes to have a uniform field of 0.01T inside a solenoid Of length 0.2m. The maximum current to be 2A. How Much wire is required to construct a suitable solenoid if The average radius of a turn is 0.01m?

Solution:

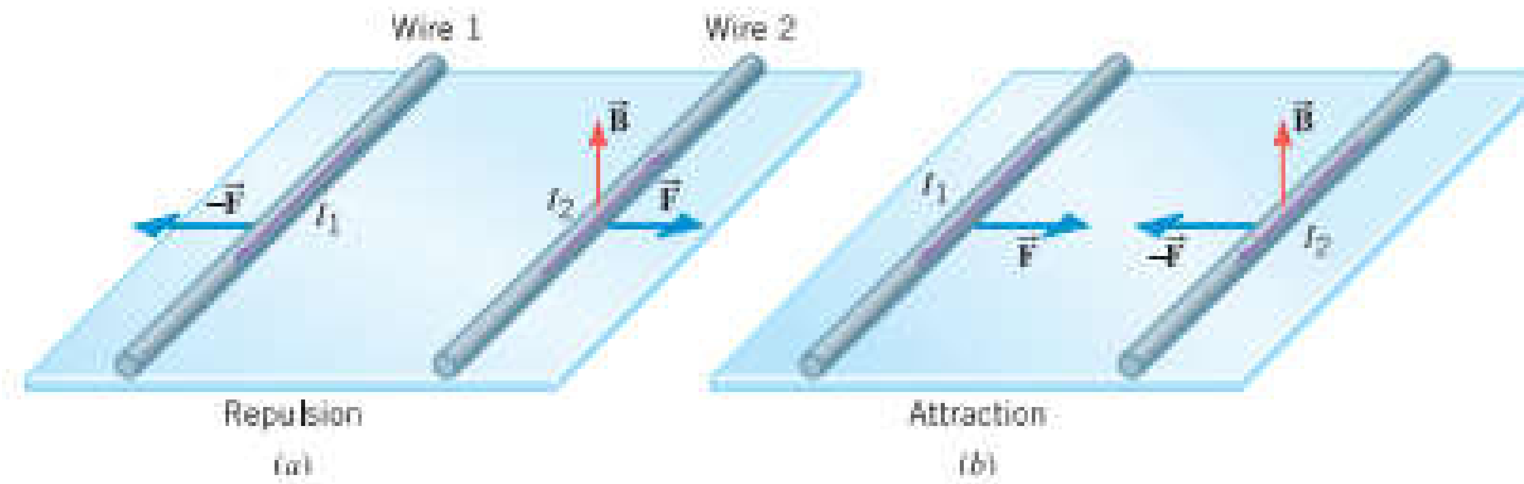
$$B = n\mu_o I = \frac{N}{L}\mu_o I \Rightarrow N = \frac{BL}{\mu_o I} = \frac{0.01 \times 0.2}{4\pi \times 10^{-7} \times 2} = 796\ turns$$

Each turn has length = $2\pi R$

So, the total length of the wire $L = 2\pi RN$

$$L = 2\pi RN = $$
$$= 2 \times 3.14 \times 0.01 \times 796$$
$$= 50m$$

19.8 The Force Between Two Parallel Currying Wires:



(a)Two long, parallel wires carrying currents I_1 and I_2 in opposite directions repel each other. (b)The wires attract each other when currents are in the same directions

$$F_{1\to 2} = I_2 L_2 B_1 \sin 90^o = I_2 L \left(\frac{\mu_o I_1}{2\pi r} \right) = F_{2\to 1}$$

EXAMPLE 19.8: Two parallel wires carrying currents of 10A and are separated by $1mm = 10^{-3}m$. What is the Magnetic force on a a 2-m long portion of wire?

Solution:

$$\frac{F}{L} = \frac{\mu_o I_1 I_2}{2\pi r} = \frac{4\pi \times 10^{-7} \times 10 \times 10}{2 \times 3.14 \times 10^{-3}} = 0.02N/m$$

$$F = 0.02 \times L = 0.02 \times 2 = 0.04N$$